

[Title, e.g. Spaced retrieval practice in workplace training: evidence from three years of quiz logs]

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Abstract

Background. Spaced retrieval practice improves retention in laboratory and educational settings, but evidence from long-running workplace programmes is scarce. **Methods.** We analysed 61,310 responses given by 187 employees to 166 multiple-choice items in 9 modules of a quiz-based microlearning programme (April 2023–June 2026). Each item was re-presented to the same employee at a fixed, non-adaptive interval (median 36 days). Correctness was modelled with logistic regression including employee and item fixed effects; descriptive estimates carry cluster-bootstrap confidence intervals. **Results.** Accuracy rose from 82.6% at the first attempt to 91.8% at the 10th and later attempts. Each doubling of the number of attempts multiplied the odds of a correct answer by 1.38 (95% CI [1.32, 1.45]; range across ten sensitivity analyses 1.29–1.50). Longer intervals since the previous attempt were associated with lower accuracy (OR 0.88 per doubling of the interval). Repeated errors concentrated on the same distractor (71.6% vs 59.6% expected). Accuracy on newly encountered items did not improve with tenure (OR 0.88 per year). **Conclusions.** Repeated retrieval in routine workplace training was accompanied by steady, item-specific gains and measurable forgetting between repetitions. Adaptive scheduling, targeted feedback on persistent misconceptions and revision of poorly functioning items could increase the programme’s effectiveness.

Keywords: spaced repetition; retrieval practice; microlearning; workplace training; learning analytics; item analysis

1 Introduction

Retrieving information from memory strengthens its later retention more than restudying it (the testing effect), and practice distributed over time is retained better than massed practice (the spacing effect) [3, 4, 24]. Spaced education, in which short multiple-choice questions are delivered repeatedly over weeks or months, improved knowledge acquisition and retention in randomised trials among medical trainees [10–12], and systematic reviews report small to moderate benefits for health professionals [17, 22].

In corporate settings, short quiz-based *microlearning* formats are widely used, but the evidence base is dominated by small or grey-literature studies [28]. Evaluations of mandatory organisational training, for example in security awareness, frequently report modest or decaying effects [9, 14, 23]. At the same time, the log data generated by such platforms allow learning to be observed at the

level of single responses, as in learner models developed for intelligent tutoring and language-learning systems [5, 13, 21, 25, 26].

[One or two sentences on the organisation and the training content.] We used three years of response logs from a workplace microlearning programme to address the following questions: (RQ1) does accuracy increase with repeated retrieval of the same item by the same employee; (RQ2) how is accuracy related to the outcome of, and the time since, the previous attempt; (RQ3) are repeated errors persistent, indicating misconceptions rather than guessing; (RQ4) does improvement extend to newly encountered items; and (RQ5) how well do the items function psychometrically?

2 Methods

2.1 Setting and training programme

[Organisation, sector, target group and training topic; whether participation was mandatory; delivery channel (e-mail, app, intranet); whether the correct answer and an explanation were shown after each response; how items were authored and reviewed.] Employees were presented with short multiple-choice questions (median 3 response options per item) grouped into thematic modules. In the observed data employees answered a median of 3 questions per active day, 100.0% of responses were given on weekdays, and each item was presented again to the same employee after a median of 36 days (IQR 18–90).

2.2 Data

The platform log contained, for each response, a record identifier, the module, the item, the option chosen, a pseudonymised employee identifier, a time stamp and the correctness of the response. Records with missing or invalid values were removed (0 records). Test module(s) (mnk-tst; 212 responses) were excluded. The analytic dataset comprised 61,310 responses (Table 1). [Ethics approval or waiver; legal basis for processing employee data (GDPR); how pseudonymisation was performed.]

2.3 Definitions

A *sequence* is the chronologically ordered series of attempts of one employee at one item; the *attempt number* is the position of a response within its sequence (ties in time were broken by record identifier). *Accuracy* is the proportion of correct responses. The *interval* is the time in days since the previous attempt in the same sequence. *Tenure* is the time since the employee’s first response in the programme. Employee-level change was defined as the difference in the proportion of correct answers between attempt 3 and attempt 1, computed across the employee’s own sequences with at least three attempts.

2.4 Statistical analysis

Descriptive accuracy is reported with 95% confidence intervals from a cluster bootstrap that resampled employees (2,000 replicates), which accounts for the dependence of responses within persons. Because learners who continue to later attempts may differ from those who stop, aggregated learning curves can be distorted by attrition [8, 18]; learning curves were therefore additionally computed for a balanced cohort of sequences with at least 5 attempts. The relationship between error rate and attempt number was summarised by a power function [20]. Because the scheduler was not adaptive, we checked whether re-presentation depended on the previous outcome.

Correctness was modelled with logistic regression including fixed effects for employee and item and with standard errors clustered by employee, so that the attempt effect is estimated within persons and within items. Model 1 included the attempt number on the $\log_2$ scale, so the odds ratio (OR) refers to a doubling of the number of attempts. Model 2, fitted to repeated attempts, added the outcome of the previous attempt, the interval since the previous attempt ($\log_2$ days) and tenure, analogous to performance-factors and half-life regression models of learner logs [21, 25]. Model 3, fitted to first attempts only, tested whether accuracy on newly encountered items changed with tenure (RQ4). Sensitivity analyses used a balanced cohort, truncation at 10 attempts, strata of sequence length (addressing pattern-mixture bias), exclusion of low-activity employees, adjustment for tenure, generalised estimating equations with an exchangeable working correlation [15], and a mixed model with crossed random intercepts for employees and items [1], from which latent-scale intraclass correlations (ICC) and marginal and conditional R^2 were derived [19].

Transitions between consecutive attempts were tabulated. For consecutive errors, the proportion of repeated choices of the same distractor was compared with the proportion expected under random choice among the observed distractors and under choice proportional to each distractor’s popularity for the item. Employee-level change was tested with the sign test and the Wilcoxon signed-rank test (rank-biserial correlation as effect size). Items were described with classical test theory indices computed on first attempts: difficulty (proportion correct), discrimination (item–rest correlation with the employee’s accuracy on the remaining items) and distractor functioning, with distractors chosen by fewer than 5% of respondents classified as non-functioning [7, 27]. Per-item trends across attempts were tested with logistic regression with employee-clustered standard errors, controlling the false discovery rate at 0.05 [2]. Analyses were performed in Python 3.14.4 with pandas 3.0.3, NumPy 2.5.0, SciPy 1.18.0, statsmodels 0.14.6 and Matplotlib 3.11.0 (random seed 2026).

3 Results

3.1 Sample and engagement

The dataset comprised 61,310 responses from 187 employees to 166 items in 9 modules, collected between April 2023 and June 2026 (Table 1). Employees gave a median of 218 responses (IQR 67–596) to 90 different items on 84 active days, and remained in the programme for a median of 18 months; 56.1% were active in the last 90 days of observation. Activity varied over time, with periods without responses (Figure 1). [Explain the gaps, e.g. programme pauses or module launches.] Overall accuracy was 86.6% and first-attempt accuracy 82.6%.

The re-presentation schedule did not depend on the previous outcome: the probability of a further attempt was 73.9% after a correct and 74.7% after an incorrect response, and the median interval to the next attempt was 36 and 38 days, respectively. Continuation of sequences was therefore not selected on performance.

3.2 Learning across repeated attempts (RQ1)

Accuracy increased steadily with the attempt number, from 82.6% [81.2, 84.0] at the first attempt to 91.8% [88.5, 94.4] at the 10th and later attempts (Figure 2, Table 2). The increase was of the same size in the balanced cohort of 5,205 sequences with at least 5 attempts (81.9% [79.8, 83.8] at attempt 1 vs 89.1% [87.3, 90.7] at attempt 5), so it is not explained by selective continuation of better-performing employees. The error rate followed a power function of the attempt number (exponent 0.28, $R^2 = 0.98$), consistent with the power law of practice [20].

Table 1: Characteristics of the dataset.

Characteristic	Value
Responses	61,310
Employees	187
Items (questions)	166
Modules	9
Observation period	2023-04-24 – 2026-06-08
Employee–item sequences	15,907
Response options per item, median (range)	3 (1–5)
Responses per employee, median [IQR]	218 [67, 596]
Items per employee, median [IQR]	90 [26, 140]
Active days per employee, median [IQR]	84 [28, 220]
Responses per active day, median [IQR]	3 [2, 4]
Responses on weekdays (%)	100.0
Months between first and last response, median [IQR]	18 [5, 30]
Employees active in the last 90 days of data (%)	56.1
Attempts per sequence, median [IQR] (max)	3 [2, 5] (19)
Sequences with ≥ 2 / ≥ 3 attempts (%)	78.6 / 60.5
Days between attempts, median [IQR]	36 [18, 90]
Correct responses, overall (%)	86.6
Correct responses, first attempts (%)	82.6

IQR, interquartile range. A sequence is the series of attempts by one employee at one item. Excluded test module(s): mnk-tst (212 responses).

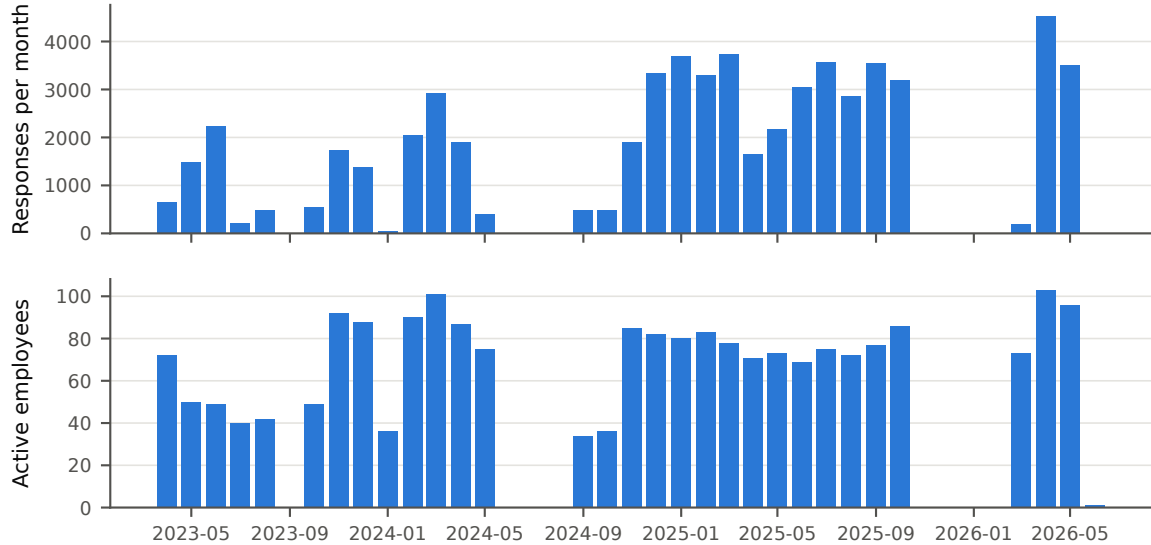


Figure 1: Monthly number of responses (top) and of active employees (bottom) over the observation period.

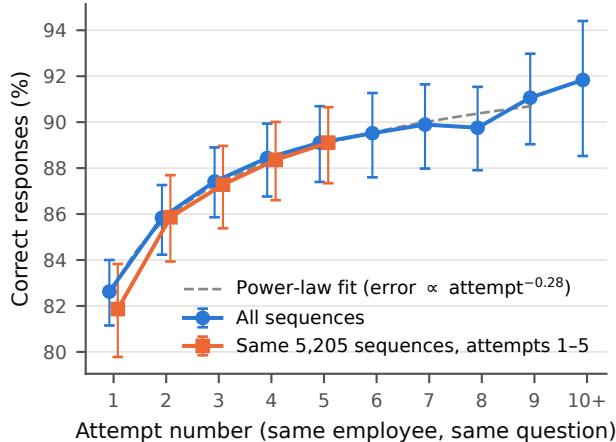


Figure 2: Accuracy by attempt number within employee–item sequences. Circles: all sequences available at each attempt (their number shrinks as attempts increase). Squares: balanced cohort, i.e. the same 5,205 sequences that reached at least 5 attempts, followed over attempts 1–5 only, so that every point is based on identical sequences. Error bars: 95% CI from a cluster bootstrap over employees (2,000 replicates). Dashed line: power-law fit to the error rate. Attempts 10 and higher are pooled.

In Model 1, each doubling of the number of attempts was associated with 1.38-fold higher odds of a correct answer (95% CI [1.32, 1.45]; Table 3). Items accounted for a larger share of the variance than employees (latent ICC 0.25 vs 0.09; marginal $R^2 = 0.021$, conditional $R^2 = 0.36$).

Table 2: Accuracy by attempt number within employee–item sequences.

Attempt	<i>n</i> Employees	%	95% CI	<i>n</i> (cohort)	% (cohort)	95% CI (cohort)
1	15,907	187	82.6 [81.2, 84.0]	5,205	81.9	[79.8, 83.8]
2	12,504	160	85.8 [84.2, 87.3]	5,205	85.9	[83.9, 87.7]
3	9,625	149	87.4 [85.9, 88.9]	5,205	87.3	[85.4, 89.0]
4	7,163	141	88.4 [86.8, 89.9]	5,205	88.4	[86.6, 90.0]
5	5,205	129	89.1 [87.4, 90.7]	5,205	89.1	[87.3, 90.7]
6	3,702	106	89.5 [87.6, 91.3]			
7	2,642	90	89.9 [88.0, 91.6]			
8	1,845	81	89.8 [87.9, 91.5]			
9	1,186	70	91.1 [89.0, 93.0]			
10+	1,531	57	91.8 [88.5, 94.4]			

Cohort: sequences with ≥ 5 attempts, which keeps the set of sequences constant across attempts 1–5 and removes survivorship bias. 95% confidence intervals from a cluster bootstrap resampling employees (2,000 replicates).

3.3 Previous outcome, interval and forgetting (RQ2)

Controlling for the attempt number, a correct previous response was associated with higher odds of a correct answer (Model 2: OR 3.85, 95% CI [3.49, 4.26]), and longer intervals since the previous attempt with lower odds (OR 0.88 per doubling of the interval, 95% CI [0.86, 0.90]; Table 3). Among previously correct responses, accuracy decreased from 95.2% when the item was repeated within 14 days to 88.5% after 240 days or more; previously incorrect responses were corrected in 61.0% and

Table 3: Logistic regression of response correctness: odds ratios [95% CI].

Predictor	Model 1	Model 2	Model 3
Attempt number (per doubling)	1.38 [1.32, 1.45]	1.33 [1.24, 1.42]	
Previous attempt correct		3.85 [3.49, 4.26]	
Days since previous attempt (per doubling)		0.88 [0.86, 0.90]	
Employee tenure (per year)		0.87 [0.77, 1.00]	0.88 [0.76, 1.01]
Sample	All attempts	Attempts ≥ 2	First attempts
Responses	60,680	44,499	15,333
Employees / items	163 / 164	146 / 163	162 / 161

All models include fixed effects for employee and item; standard errors are clustered by employee. Employees or items with no variation in correctness are dropped by the conditional likelihood (Model 1: 630 responses). Model 3 tests whether accuracy on newly encountered items improves with time in the programme.

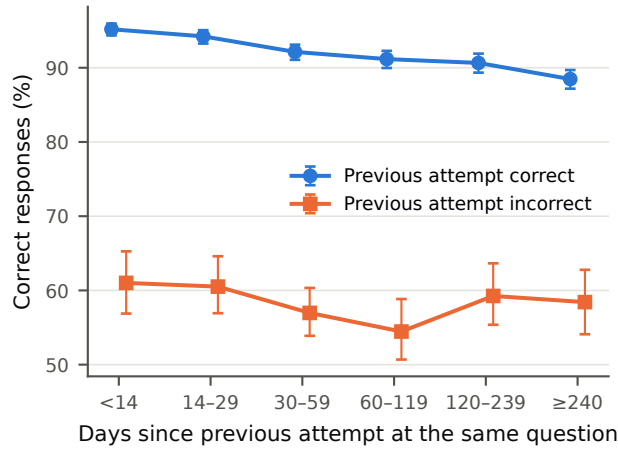


Figure 3: Accuracy by the interval since the previous attempt at the same item, separately for previously correct (circles) and previously incorrect (squares) responses. Error bars: 95% CI from a cluster bootstrap over employees.

58.4% of cases, respectively (Figure 3, Table 4). This pattern is consistent with forgetting between repetitions [3, 4].

3.4 Transitions and persistent errors (RQ3)

Of previously correct responses, 92.6% [91.8, 93.3] were correct again at the next attempt (7.4% were forgotten); of previously incorrect responses, 58.5% [55.7, 61.3] were corrected and 41.5% remained incorrect (Table 5). When an error was repeated (2,546 cases), employees chose the same distractor in 71.6% [69.4, 73.8] of cases, compared with 47.1% expected under random choice and 59.6% expected given the popularity of each distractor, indicating stable misconceptions rather than guessing.

3.5 Employee-level change

Among the 149 employees (79.7% of all) with at least one sequence of three or more attempts, 102 improved, 20 worsened and 27 did not change between the first and third attempt (sign test $p < 0.001$; Wilcoxon $p < 0.001$; rank-biserial $r = 0.72$). The mean change was 3.8 percentage points (95% CI [2.4, 5.2]; Figure 4, Table 6).

Table 4: Accuracy by interval since the previous attempt at the same item, for previously correct (left) and incorrect (right) responses.

Days since previous	n	Correct, % [95% CI]	n	Correct, % [95% CI]
<14	6,316	95.2 [94.4, 96.0]	949	61.0 [56.9, 65.3]
14–29	10,890	94.2 [93.3, 95.1]	1,631	60.5 [56.9, 64.6]
30–59	8,382	92.1 [91.1, 93.1]	1,327	57.0 [53.9, 60.3]
60–119	6,261	91.2 [90.0, 92.3]	964	54.5 [50.7, 58.8]
120–239	4,236	90.7 [89.3, 91.9]	724	59.3 [55.4, 63.7]
≥ 240	3,177	88.5 [87.2, 89.7]	546	58.4 [54.1, 62.8]

95% confidence intervals from a cluster bootstrap resampling employees (2,000 replicates).

Table 5: Transitions between consecutive attempts at the same item.

Previous attempt	n	Next correct (%)	95% CI	Next incorrect (%)
Correct	39,262	92.6	[91.8, 93.3]	7.4
Incorrect	6,141	58.5	[55.7, 61.3]	41.5

After two consecutive errors, the same distractor was chosen in 71.6% [69.4, 73.8] of cases ($n = 2,546$); the value expected under random choice among the observed distractors was 47.1%, and 59.6% when weighted by each distractor’s overall popularity for the item. 95% confidence intervals from a cluster bootstrap resampling employees (2,000 replicates).

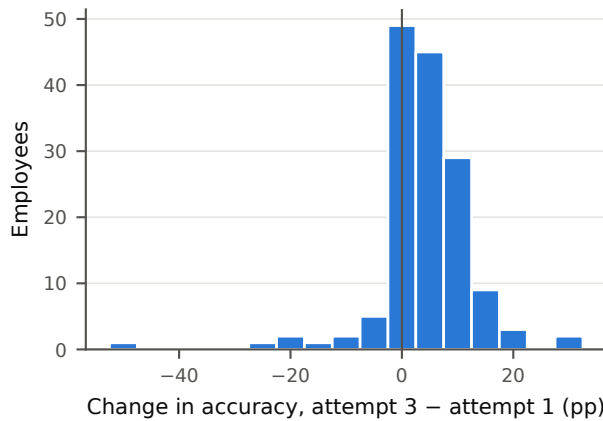


Figure 4: Distribution of the employee-level change in accuracy between attempt 3 and attempt 1 across the employee’s own sequences with at least three attempts ($n = 149$ employees; bins of 5 percentage points centred on zero).

Table 6: Change in accuracy between the first and third attempt at the same items.

Measure	Value
Employees with ≥ 1 sequence of ≥ 3 attempts	149
Improved / worsened / unchanged	102 / 20 / 27
Mean change, pp [95% CI]	3.8 [2.4, 5.2]
Median change, pp	3.7
Sign test, proportion improving [95% CI]; p	0.84 [0.76, 0.90]; <0.001
Wilcoxon signed-rank W ; p ; rank-biserial r	1,055; <0.001; 0.72
Sequences (employee-item) with ≥ 3 attempts	9,625
Correct at attempt 1 / attempt 3 (%)	82.7 / 87.4
Incorrect $\rightarrow$ correct / correct $\rightarrow$ incorrect	1,075 / 625
McNemar χ^2 ; p (ignores clustering)	118.6; <0.001

Employee-level change: difference in the proportion of correct answers between attempt 3 and attempt 1 across the employee’s own sequences with ≥ 3 attempts; ties excluded from the sign and Wilcoxon tests. pp, percentage points.

3.6 Transfer to new items (RQ4)

Accuracy at the first encounter with a new item did not increase with tenure (Model 3: OR 0.88 per year, 95% CI [0.76, 1.01]; Table 3), and tenure was not positively associated with accuracy on repeated attempts once the attempt number was controlled (Model 2: OR 0.87, 95% CI [0.77, 1.00]). The gains therefore appear to be specific to the repeated items.

3.7 Item analysis (RQ5)

The median item difficulty was 85.7% correct at the first attempt: 4 items were answered correctly by fewer than 50%, 102 by 50–90% and 60 by more than 90% of employees (Table 7). The median item–rest correlation was 0.22; 71 of 161 items had correlations below 0.20 and 48 of 0.30 or more (Figure 5). Errors concentrated on one distractor (median 71.8% of an item’s errors), and 193 of 349 observed distractors (55.3%) were non-functioning. Accuracy improved significantly with repetition for 44 of 164 testable items and deteriorated for 1 (FDR-adjusted). The most difficult items are listed in Table 8.

3.8 Sensitivity analyses

The attempt effect was stable across specifications, with ORs from 1.29–1.50 (Figure 6, Table 9): 1.36 in the balanced cohort, 1.30, 1.35 and 1.40 in sequences of 3–4, 5–7 and at least 8 attempts, 1.29 in the population-averaged GEE model and 1.37 in the mixed model. The estimate was larger after adjustment for tenure (1.50), which is correlated with the attempt number.

4 Discussion

4.1 Principal findings

In three years of routine use of a workplace microlearning programme, accuracy on repeatedly presented items rose steadily with the number of retrievals, following a power function, and the effect was robust to attrition, model specification and clustering. Accuracy declined with the interval since the previous retrieval, previous errors were only partly corrected at the next attempt, and repeated errors mostly reproduced the same distractor. Accuracy on new items did not increase with tenure.

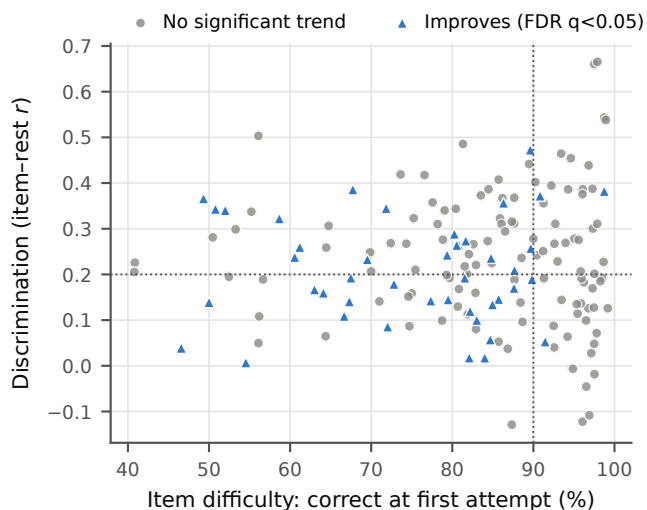


Figure 5: Item difficulty (accuracy at first attempt) versus discrimination (item–rest correlation). Triangles: items whose accuracy increased significantly with repetition (Benjamini–Hochberg $q < 0.05$). Dotted lines: $r = 0.20$ and 90% correct.

Table 7: Item analysis (classical test theory).

Measure	Value
Items analysed	166
Difficulty (first-attempt % correct), median [IQR]	85.7 [75.3, 94.5]
Items with <50% / 50–90% / >90% correct	4 / 102 / 60
Discrimination (item–rest r), median [IQR], items with $n \geq 30$	0.22 [0.14, 0.32] ($k = 161$)
Items with $r < 0.20$ / $0.20-0.29$ / ≥ 0.30	71 / 42 / 48
Share of errors on the most frequent distractor, median [IQR]	71.8 [58.4, 84.3]
Observed distractors chosen by <5% of responses (non-functioning)	193 of 349 (55.3%)
Items improving / worsening with repetition (FDR $q < 0.05$)	44 / 1 (of 164 testable)

Difficulty and discrimination computed on first attempts; item–rest correlation uses each employee’s accuracy on the remaining items (employees with ≥ 10 first attempts). Trend: per-item logistic regression of correctness on $\log_2(\text{attempt})$ with employee-clustered SE, Benjamini–Hochberg correction. Distractors never chosen are not observable in the log and are not counted.

Table 8: The 10 most difficult items.

Item	Module	n	First (%)	Repeat (%)	Top distr. (%)	r_{it}	OR trend	q
127	qo3-q5s	150	40.8	51.4	76.5	0.21	1.15	0.759
66	ier-4pm	285	40.9	42.2	55.4	0.23	0.99	0.973
261	g25-pwd	214	46.6	69.5	75.6	0.04	1.95	<0.001
140	ier-4pm	175	49.3	83.3	100.0	0.37	3.44	<0.001
305	cgn-jqc	248	50.0	75.3	43.5	0.14	2.37	<0.001
35	ier-4pm	416	50.5	65.8	71.7	0.28	1.23	0.185
160	2qx-6t6	623	50.8	67.4	82.1	0.34	1.60	<0.001
71	ier-4pm	431	52.0	73.1	73.7	0.34	1.52	0.004
206	d6s-bu8	553	52.5	60.8	54.2	0.19	1.22	0.190
132	s5m-mc2	558	53.3	67.2	82.0	0.30	1.18	0.216

First/Repeat, accuracy at first vs later attempts; Top distr., share of errors falling on the most frequent distractor; r_{it} , item-rest correlation; OR trend, odds ratio per doubling of the attempt number; q , Benjamini-Hochberg adjusted p .

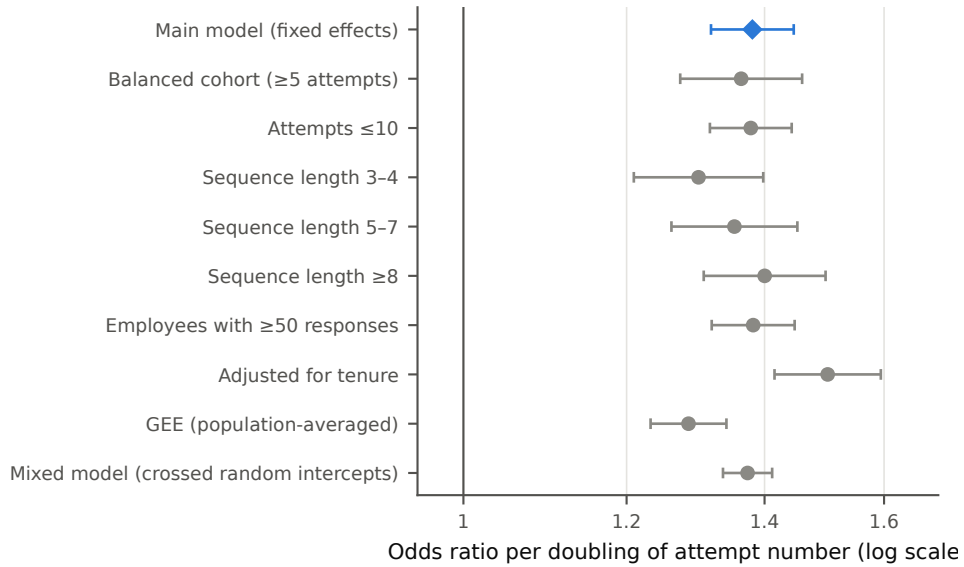


Figure 6: Odds ratio (95% CI) for a correct answer per doubling of the attempt number across model specifications. Diamond: main model; vertical line: no effect.

Table 9: Sensitivity analyses: odds ratio for correctness per doubling of the attempt number.

Analysis	Responses	OR [95% CI]
Fixed effects for employee and item (main model)	60,680	1.38 [1.32, 1.45]
Balanced cohort (sequences with ≥ 5 attempts, attempts 1–5)	24,920	1.36 [1.27, 1.46]
Attempts truncated at 10	59,877	1.38 [1.32, 1.44]
Sequences of length 3–4 only	14,218	1.30 [1.21, 1.40]
Sequences of length 5–7 only	18,154	1.35 [1.26, 1.45]
Sequences of length ≥ 8 only	16,884	1.40 [1.31, 1.50]
Employees with ≥ 50 responses	60,256	1.38 [1.32, 1.45]
Adjusted for employee tenure	60,680	1.50 [1.42, 1.59]
GEE, exchangeable correlation within employee (population-averaged)	61,310	1.29 [1.23, 1.34]
Crossed random intercepts for employee and item (Laplace)	61,310	1.37 [1.34, 1.41]

Variance of the random intercepts (logit scale): employee SD = 0.69, item SD = 1.13; latent-scale ICC: employee = 0.09, item = 0.25; Nakagawa R^2 : marginal = 0.021, conditional = 0.36. Sequence-length strata address pattern-mixture (attrition) bias in the learning curve.

4.2 Comparison with previous research

The gradual, regular gains resemble learning curves observed across educational datasets [13] and are consistent with the testing and spacing effects [3, 24]. Randomised trials of spaced education found improved retention [10, 12]; our observational data extend these findings to a non-medical workplace setting with long inter-repetition intervals. The negative association between interval and accuracy agrees with forgetting-curve models fitted to learner logs [25, 26]. As in studies that re-used the same items repeatedly [6], improvement on repeated items cannot be separated from memory for the specific question and its answer; the absence of a tenure effect on new items suggests limited transfer.

4.3 Implications for practice

The programme re-presented items at the same interval regardless of the previous outcome. Adaptive schedules that shorten the interval after errors and lengthen it after correct answers use learner history more efficiently [16, 25, 26]. The persistence of specific distractors points to misconceptions that may call for explanatory feedback rather than repetition alone. Item analysis identified candidates for revision: items with low discrimination, items answered correctly by almost everyone at the first attempt (little room for learning) and non-functioning distractors [7, 27].

4.4 Limitations

The study is observational and has no control group, so the gains cannot be attributed causally to the programme; improvement on repeated items may partly reflect recognition of the item rather than understanding. Participation and activity varied between employees, and results for sequences with many attempts describe the more active participants, although the balanced-cohort and sequence-length analyses did not suggest attrition bias. The high first-attempt accuracy (82.6%) limits the room for improvement (ceiling effect). Distractors never chosen are not visible in the log, so distractor functioning may be overestimated. No information on employees’ roles, prior knowledge or demographics was available. Time stamps carried no time-zone information. The mixed model in-

cluded random intercepts only; employee-specific learning rates were not modelled. The data come from a single organisation. [\[Add context-specific limitations.\]](#)

5 Conclusions

Routine repeated retrieval in workplace training was accompanied by steady, item-specific improvement and by forgetting between repetitions. Log data from such programmes can be used to monitor learning, evaluate items and inform the design of adaptive repetition schedules.

Declarations

Ethics. [\[Approval / waiver and data-protection basis.\]](#)

Data availability. [\[Availability of pseudonymised data and analysis code.\]](#)

Funding. [\[Funding sources.\]](#)

Competing interests. [\[Declaration.\]](#)

Use of AI tools. [\[Declaration of the use of generative AI, if required by the journal.\]](#)

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